

INDIAN KNOWLEDGE SYSTEMS (IKS)

HS271TA | Unit 1 – Structured Study Notes

Perception and prologue • Historicity of IKS • Indian Traditional Knowledge • Nature and scope • Kinds of traditional knowledge • Transition from traditional to Western knowledge • The IKS Corpus • Vedic Corpus • Indian Philosophical Systems

► UNIT AT A GLANCE

1. Perception and Prologue
2. Indian Traditional Knowledge (TK) – meaning, expression, threats, need to protect, types
3. Transition from Traditional to Western Knowledge
4. Historicity of IKS
5. The IKS Corpus – A Classification Framework (Literary / Non-Literary, Chaturdasa Vidyas)
6. Vedic Corpus – Four Vedas
7. Vedangas – Six limbs of the Vedas
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1. Perception and Prologue

Definition – Indian Knowledge System (IKS): a collection of knowledge and practices that have developed in India over **thousands of years**.

Areas covered: philosophy, mathematics, medicine (like **Ayurveda**), arts, architecture, farming, and understanding nature.

- This traditional wisdom was **passed down through generations** and has helped people live in harmony with nature and society.
- Today, in the 21st century, there is a growing interest in **bringing this ancient knowledge together with modern science and technology**. By combining the old with the new, we can create better and more sustainable ways to live.

Examples of integration:

- Traditional farming methods + modern tools → protect the environment and produce healthy food.
- Ayurveda + modern healthcare → improves overall well-being.

Colonial impact

- With the adoption of **colonial education systems**, indigenous knowledge was gradually **marginalized**.
- **Western epistemologies** came to dominate formal education, leading to a **disconnect between modern learners and their cultural roots**.

National Education Policy (NEP) 2020

- Acknowledges this gap and advocates the **revival of IKS through curriculum integration at all levels of education**.
- Emphasizes **interdisciplinary learning, inclusion of indigenous languages**, and recognition of traditional wisdom as essential for **sustainable development and cultural preservation**.
- Success of these reforms depends largely on **how students perceive and value IKS**.

✓ SECTION SUMMARY

IKS = thousands of years of Indian knowledge (philosophy, maths, Ayurveda, arts, architecture, farming). Colonial education marginalized it; NEP 2020 seeks to revive it through curriculum integration.

2. Indian Traditional Knowledge (TK)

2.1 Definition and Meaning

Traditional knowledge or **indigenous knowledge** is developed by local indigenous communities that are well-developed and whose long-standing customs and practices are followed by these communities from generation to generation.

- Studies into the origins, behaviour, and development of humanism in both prehistoric and modern societies led to the recognition of TK.
- Adoption of TK by international organisations, like the **UN**, has resulted in its widespread appreciation and recognition.

Definition – Traditional Knowledge (TK): the knowledge, know-how, skills, and practices **developed, maintained, and passed down from one generation to another**. These practices are frequently integral to the community's **cultural or spiritual identity**.

Meaning / Scope: TK can be found in a wide range of fields – **agriculture, science, technology, ecology, medicine, and biodiversity-related topics**.

- TK includes both the knowledge itself and **conventional cultural expressions**, such as signs and symbols. It comes from intellectual activity within a traditional context.
- Includes traditional use and management of **lands and resources** using indigenous agricultural methods.
- Includes **traditional medicines (herbs and spices)** and medical practices – e.g., **Ayurveda, Yoga, Unani, Siddha, and Naturopathy** (recognised in India).

2.2 Ways of Expression

- Frequently passed down **orally** from generation to generation; developed from experience gained over many years and **adapted to the local culture and environment**.
- **Belongs to everyone as a group** and is preserved through **tales, songs, folklore, proverbs, cultural values, beliefs, rituals**, etc.
- Indigenous people follow oral traditions, including **dances, paintings, carvings**, and other artistic expressions, practised and passed down through millennia.

2.3 Threats to Traditional Knowledge

TK has faced centuries of discrimination and exploitation through **colonisation** (earlier), **capitalism** (through MNCs, etc.) and ever-increasing **consumerism**.

Threat	Explanation
Language threat	Indigenous languages and dialects preserve enormous amounts of TK (e.g., Ayurveda in Sanskrit). These languages have been at the receiving end of colonial exploitation and modern lifestyles.

Threat	Explanation
External pressures	Traditional methods of preserving/transmitting knowledge are threatened by migration, environmental pressures, impact of modern lifestyles, disruption of traditional ways of life, and globalisation.
Biopiracy	Commercial exploitation of biochemicals or genetic materials that occur naturally. TK consists of many biological features and genetic diversity of the natural environment. Example: a multinational corporation's patenting of the medicinal qualities of the turmeric plant, even though it has been used for thousands of years in Indian Ayurvedic medicine.

2.4 Need to Protect Traditional Knowledge

Preservation of TK is crucial because it plays a crucial role in the **social and physical environment** of a community.

Reason	Explanation
Protecting Indigenous culture and identities	TK and languages are a significant way to maintain and preserve Indigenous cultures and identities and promote well-being .
Protecting livelihoods	TK is one of the sources of livelihood for indigenous people.
Health benefits	TK is connected with environment and spirituality, so it is important to well-being. Traditional medicines can provide health benefits to a large population, like in India.
Ecological benefits	Opportunities to conserve forests and biodiversity . Example: sacred groves / forest temples conserve biodiversity.

2.5 Different Types of Traditional Knowledge

#	Type of Traditional Knowledge
1	Cultural Knowledge
2	Artistic Knowledge
3	Medicinal Knowledge
4	Biodiversity / Natural Resources Knowledge
5	Agricultural Knowledge
6	Sacred Knowledge

✓ SECTION SUMMARY

TK = knowledge, know-how, skills, practices passed down generations; mostly **oral**, community-owned.

Threats: **language loss, external pressures, biopiracy** (turmeric patent).

6 types: Cultural, Artistic, Medicinal, Biodiversity/Natural Resources, Agricultural, Sacred.

3. Transition from Traditional to Western Knowledge

The transition in India marked a **profound shift** in the way learning was conceived, transmitted, and institutionalized.

Traditional Indian education

- Rooted in the **Gurukula system** and temple-based learning centers such as **Nalanda and Takshashila**.
- Emphasized **holistic development** – blending philosophy, science, arts, and ethics within the framework of **Sanatana Dharma**.
- Knowledge was viewed as sacred (**Vidya**), aimed at **self-realization and societal harmony rather than material gain**.
- **Oral traditions, Sanskrit texts, and regional commentaries** ensured continuity of indigenous sciences – Ayurveda, astronomy, mathematics, metallurgy, and linguistics.

Westernization under British colonial rule

- Introduction of the **English language, new curricula, and modern institutions**.
- Focus shifted from **spiritual and moral education** to **empirical, utilitarian, and job-oriented learning**.
- Created a **divide** between traditional knowledge systems and Western scientific paradigms.
- However, it also facilitated **documentation and global exposure** of Indian heritage, paving the way for a **contemporary synthesis** – modern education increasingly seeks to reintegrate IKS with modern scientific inquiry and innovation.

Aspect	Traditional (Indian) Education	Western (Colonial) Education
Institutions	Gurukula, Nalanda, Takshashila	Modern institutions, new curricula
Language	Sanskrit texts, regional commentaries, oral tradition	English language
View of knowledge	Sacred (Vidya)	Empirical, utilitarian
Aim	Self-realization and societal harmony	Job-oriented / material gain
Approach	Holistic – philosophy, science, arts, ethics	Divided disciplines; secular, empirical

★ KEY POINT

Traditional aim = **self-realization**; Western (colonial) aim = **job-oriented learning**. Today's goal = **synthesis** of IKS and modern science.

4. Historicity of Indian Knowledge Systems

IKS finds its origins in the **Vedic period (circa 1500–500 BCE)**, marked by the compilation of the **Vedas**, the foundational texts of Indian civilization.

- **Rigveda, Yajurveda, Samaveda, Atharvaveda** encompass cosmology, spirituality, and ethics.
- Supplementary texts like the **Upanishads** delve into metaphysics – the nature of the self (**Atman**) and ultimate reality (**Brahman**).

4.1 Epics and Classical Texts

- **Ramayana and Mahabharata** – epic narratives and repositories of moral and ethical guidance.
- **Bhagavad Gita** (part of the Mahabharata) – a philosophical treatise that continues to inspire scholars and practitioners globally.
- **Kalidasa's Shakuntala** – literature; **Patanjali's Yoga Sutras** – holistic well-being.
- **Philosophy**: Six classical schools – **Nyaya, Vaisheshika, Samkhya, Yoga, Mimamsa, Vedanta** – form the foundation of intellectual discourse (logic, metaphysics, spirituality).

4.2 Scientific and Mathematical Contributions

Field	Contributors / Works	Contribution
Mathematics	Aryabhata, Brahmagupta, Bhaskaracharya	Pioneered zero, decimal system, advanced trigonometry . Aryabhata's work <i>Aryabhatiya</i> introduced the concept of Earth's rotation and revolution ; Brahmagupta formulated rules for solving quadratic equations .
Astronomy	Varahamihira – <i>Brihat Samhita</i> ; Aryabhata	Sophisticated models to predict celestial movements. Brihat Samhita combined astronomy with meteorology and astrology; Aryabhata accurately calculated the Earth's circumference .
Medicine	Sushruta – <i>Sushruta Samhita</i> ; Charaka – <i>Charaka Samhita</i>	Ayurveda – holistic healing through herbal remedies, dietary practices, therapeutic techniques. Sushruta Samhita = seminal text in surgery ; Charaka Samhita = internal medicine .

4.3 Ancient Libraries and Educational Institutions

Dissemination and preservation of knowledge were integral to Indian civilization; institutions and libraries became centers of learning, attracting scholars from across the world.

Institution	Details
Nalanda University	Established around the 5th century CE ; a global hub of education with a vast library (Dharmaganja) on a wide range of subjects. Hosted students and scholars from China, Korea, and Central Asia .
Takshashila	One of the earliest universities (circa 600 BCE); offered education in medicine, law, and military science .

Libraries and Manuscripts:

- Manuscripts written on **palm leaves and birch bark** were meticulously preserved in temples, monasteries, and private collections.
- Scripts like **Brahmi, Kharosthi, and later Devanagari** played crucial roles in documenting knowledge.
- **Architectural texts** such as **Manasara, Mayamata, and Samarangana-sutradhara** also included science and technology.

✓ SECTION SUMMARY

Origin: Vedic period (c. 1500–500 BCE). Epics = moral guidance; Gita = philosophy.

Maths: zero, decimal, trigonometry. Astronomy: Aryabhata, Varahamihira. Medicine: Charaka (medicine), Sushruta (surgery).

Nalanda (5th c. CE, Dharmaganja library) and Takshashila (c. 600 BCE) were global learning centers.

5. The IKS Corpus – A Classification Framework

The vast corpus of IKS can be broadly classified into two parts: the **Literary Part** and the **Non-Literary Part**. Together they represent the holistic nature of India's intellectual, scientific, and cultural heritage, encompassing **both written and oral traditions**.

Level 1	Level 2	Level 3 / Contents
IKS – Literary	Sanatana Dharma	Core: 14 Vidyasthanas; works of several religious leaders and philosophers Other: Basic & applied sciences; Engineering, Technology, Architecture, Alchemy; Aesthetics, Kavyas, Arts; Health, Wellness, Psychology; Public Administration; Code of Living, etc.
	Other Dharmik Traditions	Buddhist, Jain
	Regional	Wealth of religious and other literature in all major Indian languages
IKS – Non-Literary	Oral Traditions	64 Kalas; Art forms; Health; Food & Life Practices; Folklore

5.1 The Literary Part

The textual and written body of knowledge. Divided into two major components: the **Sanatana Dharma Corpus** and **Other Dharmik Literature**, covering philosophical, scientific, and artistic disciplines.

(a) The Sanatana Dharma Corpus

- Forms the **foundational framework** of Indian civilization.
- **Core scriptures:** the **Vedas and Vedangas** – primary building blocks of Sanatana Dharma and the ultimate sources of sacred and scientific wisdom.
- Supplemented by literature on **mathematics, astronomy, aesthetics, engineering, and technology**.
- **Example:** mathematical foundations used in astronomical calculations, particularly those related to **Yuga cycles**, show how scientific inquiry was interwoven with metaphysical concepts.
- Most works were composed in **Sanskrit**; later regional languages retold, adapted, or contextualized these ideas for broader audiences.

(b) Other Dharmik Literature

Buddhist and Jain literature – vast bodies of knowledge, particularly between **500 BCE and the early centuries CE**, contributing to mathematics, natural sciences, and philosophy.

Tradition	Contribution
Buddhist	Incorporated mathematical concepts; extended into practical sciences such as maritime activity, shipbuilding, and alchemy . Example: Rasaratnakara by Nagarjuna (1st century CE) – one of the earliest treatises on alchemy.
Jain	Canonical sacred texts integrate mathematics as a core discipline (Ganitanuyoga) . Works: Tattvartha Sutra ; later mathematical explorations by scholars like Madhavacharya – attempts to quantify complex natural phenomena, e.g., estimating the number of species as 2⁹⁶ .

5.2 The Non-Literary Part

- Comprises **oral traditions and unwritten knowledge systems** transmitted through generations.
- Includes **health traditions, performing arts**, and the celebrated **64 Kalas** – classical art forms encompassing music, dance, painting, sculpture, craftsmanship, etc.
- Highlights the **dynamic, lived aspect** of Indian knowledge, ensuring continuity and adaptability over millennia.

5.3 Sanatana Dharma and the Sciences

- **Basic and applied sciences:** mathematics, astronomy, **plant sciences (Vriksha Ayurveda)**, engineering, technology – metalworking, shipbuilding, dam construction, watershed management.
- Knowledge of **alchemy, cosmetics, perfumes, dyes, town planning, and architecture** – sophisticated understanding of material and environmental sciences.
- **Health and wellness:** especially **Ayurveda** – *Charaka Samhita, Sushruta Samhita, Ashtanga Hridaya*.
- **Philosophical systems** – Yoga, Sankhya, Upanishads, Vedanta – explore **psychology, consciousness, and mental well-being** (science + spirituality).
- **Nitisastras** – guides to ethical conduct and governance:
 - **Raja-niti** – principles of public administration and governance
 - **Samanya-niti** – codes of ethics and morality
- Also celebrates **aesthetics, poetry (Kavyas), and performing arts**.

5.4 Chaturdasa Vidyas vs. Modern Subjects

Modern Subject	Chaturdasa Vidyas
Mathematics	Jyotisha and Kalpa
Science	Atharva Veda – Physics Ayurveda – Botany, Biology, Zoology, Chemistry
Economics	Arthashastra
Architecture	Jyotisha and Kalpa (building fire altars for Vedic sacrifices); Agama Shastra as well
Astronomy	Jyotisha – a combination of Mathematics, Astronomy and Astrology
History, Geography	Puranas

Modern Subject	Chaturdasa Vidyas
Languages	Siksha – Pronunciation Vyakarana – Grammar Chandas – Prosody Nirukta – Etymology of the Vedic words All used in learning Sanskrit , relevant even today – 'the basis for Computers!' (as per source)
Art	Shilpa Shastra and Agama Shastra
Music	Gandharvaveda
Sports	Dhanur Veda
<i>And many more...</i>	

✓ SECTION SUMMARY

IKS Corpus = **Literary** (Sanatana Dharma, Other Dharmik, Regional) + **Non-Literary** (Oral traditions, 64 Kalas).

Core of Sanatana Dharma literature = **Vedas + Vedangas**.

Nitisastra = Raja-niti (governance) + Samanya-niti (ethics).

6. Vedic Corpus

6.1 Meaning of 'Veda'

Definition – Veda: means '**Knowledge**'. A Sanskrit word from the root '**Vid**', meaning **finding, knowing, acquiring, or understanding**. The term Veda is the treasure house of all knowledge and all Sastras – an **eternal asset of India**.

- The root **vid** has **four meanings and four types of formation**:

Root meaning	Sense
Jnana (vetti)	Knowing
Vicarana (vinte)	Reflecting / considering
Sattayam (vidyate)	Being / existing
Labhe (vindati / vindate)	Obtaining / acquiring

Verse (as given): "*vetti veda vida jnane vinte vida vicarane / vidyate vida sattayam labhe vindati vindate //*"

Sayana says: "*vidanti jananti vidyante bhavanti vinte vicarayati vindate labhante sarve manusyah sattvavidyam yairyesu tatha vidvamsasca bhavanti te vedah.*"

- The ideas, teachings, and practices of the Vedas formed the basis for the **six major schools of Hindu philosophy** – Nyaya, Vaisheshika, Samkhya, Yoga, Mimamsa, and Vedanta.

- Written form dates back to about **1600 BCE**. The **Rig Veda** is the oldest of the 4 Vedas (authored in and around 1600 BCE). No definite date can be ascribed because transmission was by a **precise and elaborate oral tradition**.

6.2 The Four Vedas (Chaturveda)

There are four Vedas: **Rig Veda, Yajur Veda, Sama Veda, and Atharva Veda** – together called '**Chaturveda**'. The Rig Veda serves as the **principal one**; the other three agree with one another in form, language, and content [source says 'Arthaveda' – read as Atharva Veda].

Four portions of each Veda SABU

Portion	Description
Samhitas	The most ancient layer of text – mantras, hymns, prayers, and benedictions.
Aranyakas	Constitute the philosophy behind the ritual sacrifice .
Brahmanas	Commentary on hymns of the four Vedas.
Upanishads	Conversations between teachers and students ; clarify the Vedas' philosophical message.

6.3 Rig Veda

- One of the **oldest texts of the Indo-Aryan civilization still extant**; an ancient collection of Vedic hymns.
- Rig** = 'praise or shine'; **Veda** = 'knowledge'.
- 1,028 hymns and 10,552 verses**, organized into **ten Mandalas** (books). Each Mandala has many **Anuvaks**, and each Anuvak has many **Suktas** (hymns – group of verses) praising deities.
- Date**: around **1600 BCE** (from variation in form of Sanskrit); wider approximation **1700–1100 BCE**. Initial written Rig Veda dates to the 1st millennium BCE; extant manuscripts date only to the **11th–14th centuries** because palm leaves/birch bark are ephemeral.
- Themes**: cosmology, mystic forces, existence of the Universe, and other metaphysical issues – central theme: '*not about what exists, but about what it is to exist.*'
- Some hymns are still used in major rites of passage ceremonies in contemporary Hinduism.

Important Suktas: Purusha Sukta (Mandala 10, Sukta 90) • Nasadiya Sukta (10.129) • Hiranyagarbha Sukta (10.121) • Saraswati Sukta • Vak (Devi) Sukta • Bhoo Sukta • Shree Sukta

Shakhas: Rig Veda is said to have had **21 Shakhas; only five survived** – Shakala, Bashkala, Aashvalayana, Shankhayana, Mandukayana. The **two currently popular/surviving are Shakala and Bashkala**.

Shakha	Brahmana	Aranyaka	Upanishad
Shakala	Aitareya Brahmana	Aitareya Aranyaka	Aitareya Upanishad – contains Mahavakya ' Prajnanam Brahma '
Bashkala	Kaushitaki Brahmana	Kaushitaki Aranyaka	Kaushitaki Upanishad

6.4 Yajur Veda

- Yajus** = prose mantras dedicated to religious reverence/veneration; **Veda** = knowledge.

- Known as the '**book of rituals**' – a compilation of ritual-offering formulas / prose mantras chanted or muttered repeatedly by a priest while the individual performs ritual actions before the **sacrificial fire (Yajna)**.
- [Source states both 'second of the four Vedas' and 'third of the fourth canonical texts'; traditional order: Rig, Yajur, Sama, Atharva – i.e., second.]

Two branches: Shukla Yajur Veda and Krishna Yajur Veda. Every Veda has two components: Mantra and Brahmana (Upanishads are part of Brahmana).

Feature	Shukla Yajur Veda	Krishna Yajur Veda
Meaning	Shukla = 'brightness'	Krishna = 'dark'
Mantra & Brahmana	Separated	Given together (intertwined)
Difficulty	Relatively easier to grasp	Relatively more challenging to decipher
Surviving Shakhas	2 – Kanvya, Madhyandini	4 – Taittiriya, Maitrayani, Katha, Kapishtala

- Yajur Veda once had **101 Shakhas**; presently only **six** exist. **Sayana Acharya** explained these matters in his treatise on Shukla Yajur Veda (proficient in both branches).

6.5 Sama Veda

- '**Book of Songs**' – *Saman* = song, *Veda* = knowledge. Also the Veda of **Melodies and Chants**; third of the four Vedas.
- Words of the Rig Veda are **put to music** – to be **sung** rather than just read or recited.
- Shortest** of the four Vedas – **1549 verses**.
- Principal root of **classical Indian music and dance** tradition (claimed to be the oldest in the world).
- Two Upanishads embedded: **Chandogya** and **Kena**. (*Upanishads = essence of the Vedas; central philosophical concepts shared with Buddhism and Jainism.*)
- Three recensions survive** (Patanjali: originally 1000): **Kauthuma, Jaiminiya, Ranayaniya**.

Structure: Part I – **Gana** (melodies), separated into **Gramageya** and **Aranyageya**; Part II – **Archika** (three-verse book), split into **Purvarcika** and **Uttararcika**. A verse in the arcika books relates to a tune in the song books.

6.6 Atharva Veda

- Fourth and final Veda; 'knowledge storehouse of **Atharvanas**' – formulas and spells to counteract diseases and calamities, or '**the procedures for everyday life**'.
- Called '**the Veda of Magic formulas**'; a **late addition**; sides with popular culture and tradition rather than preaching religious and spiritual teachings; often viewed as a **discrete scripture**.
- 730 hymns, about 6,000 mantras, 20 books**; three Upanishads embedded: **Mundaka, Mandukya, Prashna**. A considerable part is an adaptation of the Rig Veda.
- Contains **surgical and medical speculations**, mantras for various ailments. Example: **hymn 4.15 (Pippalada version)** – dealing with an **open fracture** and wrapping the wound with the **Rohini plant** (*Ficus infectoria*, native to India).
- Also speculations on herbal remedies, nature of man, life, good and evil, and even spells/prayers to gain a lover.

6.7 Comparison of the Four Vedas

Feature	Rig Veda	Yajur Veda	Sama Veda	Atharva Veda
Meaning	Praise / shine	Prose mantras (ritual)	Songs / melodies	Formulas & spells for everyday life
Nature	Hymns of praise; oldest, principal	'Book of rituals'	'Book of Songs'	'Veda of Magic formulas'
Size	1,028 hymns; 10,552 verses; 10 Mandalas	—	1,549 verses (shortest)	730 hymns; ~6,000 mantras; 20 books
Shakhas surviving	5 of 21 (Shakala, Bashkala popular)	6 of 101	3 of 1000	—
Upanishads	Aitareya, Kaushitaki	—	Chandogya, Kena	Mundaka, Mandukya, Prashna

♦ MEMORY TRICK

Order of Vedas: **R-Y-S-A** → Rig, Yajur, Sama, Atharva. Sama = **S**ong, Yajur = **Y**ajna (ritual), Atharva = **A**ilments / everyday life.

✓ SECTION SUMMARY

Veda = knowledge (root *vid*). **4 Vedas**, each with **Samhita, Brahmana, Aranyaka, Upanishad**.

Rig = oldest/principal; Yajur = rituals (Shukla/Krishna); Sama = music; Atharva = everyday life, medicine, magic formulas.

7. Vedangas

Definition – Vedangas: literally mean the **limbs of the Vedas**. They are **six** in number and perform supportive and augmenting functions in the **study, preservation and protection** of the Vedas and Vedic traditions.

Six Vedangas: Siksha, Chhanda (Chandas), Vyakarana, Nirukta, Jyotisha and Kalpa.

- Collectively furnish the essential tools and knowledge for accurate **interpretation, recitation, and application** of the Vedas in religious and ritual practices – the supportive pillars of Vedic scholarship.

Vedanga	Focus	Body part (Veda Purusha)
Chandas	Vedic verse structures and meters	Legs (feet)
Kalpa	Guidelines for Vedic rituals and ceremonies	Hands
Jyotisha	Astronomy and astrology	Eyes
Nirukta	Meanings / etymology of difficult Vedic words	Ears
Siksha	Correct pronunciation of Vedic hymns	Nose
Vyakarana	Grammar – aids in understanding and structuring Sanskrit	Face (mouth)

Shloka (as given): "Chandah padau tu vedasya hastau kalpotha pathyate / jyotisham ayanam chakshur niruktam shrotram uchyate // Siksha ghranam tu vedasya mukham vyakaranam smrtam / tasmad sangam adhityaiva brahmaloke mahiyate"

♦ MEMORY TRICK

Bottom-to-top: **C**handas-Legs, **K**alpa-Hands, **J**yotisha-Eyes, **N**irukta-Ears, **S**iksha-Nose, **V**yakarana-Face (follows the shloka order: Chandas → Kalpa → Jyotisha → Nirukta → Siksha → Vyakarana; think *feet* → *hands* → *eyes* → *ears* → *nose* → *face*).

7.1 Difference between Vedas and Vedangas

Basis	Vedas	Vedangas
Purpose	Holy texts that teach about the origin of the world and reason for existence	Ancillary studies that help maintain the integrity of the Vedic tradition
Structure	Four collections – Rig, Sama, Yajur, Atharva	Six – Siksha, Chhanda, Vyakarana, Nirukta, Jyotisha, Kalpa
Content	Hymns, incantations, and rituals	Insights into meters, grammar, and linguistic analysis
Origin	Among the world's oldest religious scriptures; archaic Sanskrit; Indo-European-speaking peoples in NW India, c. 1500–1200 BCE	The last treatises of Vedic literature

7.2 Siksha (Phonetics) – 'Nose'

- Driven by the **phonetics and phonology** of pronunciation, as a process of internalization. Means '**Instruction in Reciting**' – correct pronunciation, accentuation, etc.
- Nose enables breathing and sustains life forces → Siksha is the **life breath of Vedic Mantras**.
- Phonetics is most important in Vedic language because **any change in sound leads to changes in results and effects** (*Svaravarnadi ucchaarana prakasho yatropadishyate sa shikshaa* – Sayanacharya).

Shikshavalli chapters: **Varna** Shuddhi (syllable purity), **Swara** Shuddhi (tonal purity), **Matra** Shuddhi (durational purity), **Balam** (force of articulation), **Saama** (evenness), **Santaana** (continuity).

7.3 Kalpa (Ritual) – 'Hands'

- Deals with aspects **inducing a person into Vedic action**; mastery of the other five Vedangas aims at performing Kalpa actions successfully.

It is like a **manual** giving details on:

- How a ritual should be done;
- Functions of brahmacharis, grihastas, sannyasis, etc.;
- What ritual involves which mantra, devata, materials;
- How many priests should perform a given ritual;
- What vessel of what shape and size to be used, etc.

- Kalpa Sastra (Sutra) compiled by many sages – **Apasthamba, Bodhayana, Vaikhanasa**, and others.

Four types of Kalpasutras SGDS

Kalpasutra	Deals with
Shrauta-sutras	Shrauta sacrifices
Grihya-sutras	Domestic ceremonies
Dharma-sutras	Religious and social laws
Shulba-sutras	Rules of measurement of fire-altars, etc. (geometry)

7.4 Vyakarana (Grammar) – 'Face / Mouth'

- Analysis of **grammar and linguistics**; helps in better construction of words and sentences; Vedas regard Vyakarana as the **mouth of the Vedas**.
- Also known as **analysis and derivation** – grammatically describing Sanskrit.
- Earliest examples: **Panini's grammar (c. 400 BCE)** and the **Pratishakhyas**.
- **Maheswara Sutras** – fourteen sutras, foundational to Sanskrit grammar; believed to have originated from the rhythmic sounds of **Nataraja's Damaru (drum)**.

Panini's Ashtadhyayi – monumental work with **four parts**:

Part	Description
Sivasutra (Maheswara sutra)	Phonology – notations for phonemes in fourteen succinct lines; groundwork for the phonetic structure of Sanskrit.
Ashtadhyayi	Core of Panini's grammar – morphology ; intricate construction rules for forming complex words and sentences; systematic framework for analyzing Sanskrit sentence structure.
Dhatupatha	List of roots ; classifies verbal roots by semantic and grammatical functions – building blocks of Sanskrit verbs.
Ganapatha	List of classes of primitive nominal stems ; insight into forms and declensions of nouns; aids identification of nominal formations in Vedic texts.

- Commentary on Vyakarana = **Mahabhashya** by Sage **Patanjali**; another commentary by **Vararuchi**.

7.5 Nirukta (Etymology) – 'Ears'

- Dedicated to **interpretation of Vedic words and their etymology**; 'Ears of Veda Purusha'.
- Aids understanding of **symbolic and metaphorical aspects** by unraveling deeper meanings in the language of the scriptures; reveals cultural, philosophical, and religious concepts.
- **Nighantu** – a collection (lexicon/glossary) of difficult Vedic words compiled by sage **Yaska (c. 600 BCE)**. Yaska's commentary is the most prominent and widely studied.

★ KEY POINT

Fundamental principle of Nirukta: **all names originated from verbs** – showing the link between language, thought, and reality.

7.6 Chandas (Meter) – 'Legs'

- Essential to Vedic poetry; measures and divides mantras by the **number of padas ('feet')** in a verse; each pada is divided by its number of **syllables** – the basis of a taxonomy of Vedic meter.

Meter	Padas	Syllables per pada	Total syllables
Gayatri	3	8	24
Anushtup	4	8	32 – the typical shloka of classical Sanskrit poetry

7.7 Jyotisha (Astronomy & Astrology) – 'Eyes'

- Study of **astronomy and astrology** as related to Vedic texts; insight into cosmic phenomena and their impact on human life.
- Varahamihira** – authored **Pancha Siddhantika** (five systems): **Pitamaha, Vasishta, Romaka, Paulisha, Surya Siddhanta**.
- Bhaskara Acharya** (12th century AD) – described the **Panchanga**, still widely used for auspicious timings and religious/cultural events.

Panchanga element	Meaning
Tithi	Lunar phase
Vaara	Day of the week
Nakshatra	Lunar mansion
Yoga	Auspicious combination
Karana	Half of a Tithi

✓ SECTION SUMMARY

6 Vedangas = Siksha, Chandas, Vyakarana, Nirukta, Jyotisha, Kalpa – the limbs of Veda Purusha.

Panini = Ashtadhyayi (c. 400 BCE); Yaska = Nighantu/Nirukta (c. 600 BCE); Varahamihira = Pancha Siddhantika; Panchanga has 5 elements.

8. Indian Philosophical Schools

Indian philosophical thought can be traced back to the **first available literature of the world – the Rigveda**. Hymns like **Nasadiya, Purusha, Vak, and Jnana Suktas** symbolically narrate the creation of the universe, nature of self, etc. Philosophical discourse flourishes in the **Upanishads**, the last major part of Vedic literature.

In the **Post-Vedic period**, thoughts turned into independent schools: Sankhya, Yoga, Nyaya, Vaisheshika, Mimamsa, Vedanta, Charvaka, Jaina, and Bauddha. Some carried forward Vedic thought; some opposed the validity of the Vedas.

Category	Meaning	Schools
Astika	Accepts the validity of the Vedas as a source of knowledge	Samkhya, Yoga, Nyaya, Vaisheshika, Mimamsa, Vedanta (6)
Nastika	Denies the validity of the Vedas as a source of knowledge	Charvaka, Bauddha (Buddhism), Jaina (Jainism)

#	Philosopher	System (Darshana)	Text
1	Brihaspati	Charvaka-darsana	Brihaspatya-sutra
2	Mahavira	Jaina-darsana	Agama sahitya
3	Siddhartha Gautama	Bauddha-darsana	Buddha Tripitaka
4	Kapila	Sankhya-darsana	Sankhya-sutra
5	Patanjali	Yoga-darsana	Yoga-sutra
6	Kanada	Vaisheshika-darsana	Vaisheshika-sutra
7	Gautama	Nyaya-darsana	Nyaya-sutra
8	Jaimini	Mimamsa-darsana	Mimamsa-sutra
9	Badarayana	Vedanta-darsana	Vedanta-sutra

School	Core idea
Nyaya	Logic & analytic philosophy
Vaisheshika	Few atomic building blocks & consciousness
Mimamsa	Critical interpretation of the Vedas
Vedanta	The 'essence' of the Vedas
Samkhya	Consciousness & Matter
Yoga	Meditation, Contemplation

8.1 Nastika Schools (Non-Vedic)

(a) Charvaka

- **First and foremost nastika darshana**; called **Lokayata** ('which appeals to the mass'); attributed to **Brihaspati** or his disciple (hence **Barhaspatya darshana**).
- Chanakya in his *Arthashastra* calls Brihaspati the foremost teacher of Arthashastra. Believed to be as old as the Vedic tradition.
- **Extreme nastika** – only **one valid means of knowledge: direct perception (pratyaksha)**; all other sources are untrustworthy or misleading.
- Not any supernatural power is god – the **king** who has the power of punishing or rewarding people should be considered god, since we know him by direct perception.

(b) Buddhist Philosophy

- Seed in the teachings of **Gautama Buddha (earlier name Siddhartha)**; he emphasised a **moral life for emancipation from human suffering** rather than philosophical problems. Later scholars developed a profound philosophy.
- Non-theistic** – not especially concerned with affirming or denying a creator God.
- Rejects blind faith** in the Vedas or other revealed scriptures.
- Central doctrines: **Four Noble Truths** and the **Noble Eightfold Path** – aim at ending suffering (**dukkha**).

(c) Jain Philosophy

- Based on the teachings of **24 Tirthankaras**; **Rishabhadeva** is the first. Last two – **Parshvanatha and Mahavira (6th century BC)** – are historical personalities.
- Anekantavada** (pluralism / multiple viewpoints) – reality is complex and can't be captured fully by any single perspective.
- Emphasizes **non-violence (ahimsa)**, spiritual purity, self-control, and liberation through ethical and ascetic practice.
- Only the **Kevalis** (those with infinite knowledge) know the full truth; others perceive only parts of it.

8.2 Vedic (Astika) Philosophical Systems

School	Founder / Text	Key Ideas
Samkhya	Kapila – Sankhya-sutra	Oldest of the orthodox systems. Everything stems from Purusha (self/soul/mind) and Prakriti (matter, creative agency, energy). Dualist – between self and matter (not mind and body as in Western dualism). Liberation = realization that soul and the gunas (steadiness, activity, dullness) are different.
Yoga	Patanjali – Yoga Sutras (2nd c. BC, as per notes)	Accepts Samkhya psychology and metaphysics but is more theistic (adds a divine entity to Samkhya's 25 elements). Yoga Sutras divided into eight Ashtanga (limbs) – like Buddhism's Eightfold Path. Goal: quiet the mind and achieve kaivalya (solitariness/detachment).
Nyaya	Aksapada Gautama – Nyaya Sutras (2nd c. BC)	System of logic , influential for many Indian schools. Valid knowledge (four sources: perception, inference, comparison, testimony) is essential to free oneself from suffering; develops criteria to judge knowledge valid/invalid.
Vaisheshika	Kanada (6th c. BC)	Atomist and pluralist – all physical objects are composed of atoms ; Brahman (metaphysically) is the force causing consciousness in atoms. Originally accepted only two sources: perception and inference . Later merged with Nyaya (close metaphysical views).
Purva Mimamsa	Jaimini	Objective: interpret and uphold authority of the Vedas ; insists on unquestioning faith in the Vedas and ritual Vedic fire sacrifices to maintain cosmic order. Salvation must be earned through action per Vedic prescriptions . Later views included Brahman and liberation.

School	Founder / Text	Key Ideas
Vedanta (Uttara Mimamsa)	Badarayana – Vedanta-sutra	Focuses on the Upanishadic/spiritual portion of the Vedas (not ritualistic Brahmanas); stresses meditation, self-discipline, spiritual connection over ritualism. Has major sub-schools (below).

Vedanta sub-schools

Sub-school	Position
1. Advaita (non-dualism)	Individual soul (Atman) and Brahman are ultimately one.
2. Visishtadvaita	Supreme Being (Vishnu) has definite attributes; world and individual souls are distinct yet inseparable modes of Brahman.
3. Dvaita	Three separate realities: Vishnu, the eternal soul, and matter .

[Note: the source says Vedanta has six sub-schools but lists only three. The remaining commonly cited ones are Dvaitadvaita, Shuddhadvaita, and Achintya Bhedabheda – add from your class notes/textbook if required.]

♦ MEMORY TRICK

Astika (6): S-Y-N-V-M-V → Samkhya, Yoga, Nyaya, Vaisheshika, Mimamsa, Vedanta.

Nastika (3): C-B-J → Charvaka, Bauddha, Jaina.

✓ SECTION SUMMARY

Astika = accept Vedas (6 schools); Nastika = reject Vedas (Charvaka, Buddhism, Jainism).

Charvaka → only perception; Buddhism → Four Noble Truths; Jainism → Anekantavada + Ahimsa.

Samkhya = Purusha + Prakriti; Yoga = 8 limbs, kaivalya; Nyaya = 4 pramanas; Vaisheshika = atoms;

Mimamsa = Vedic ritual; Vedanta = Upanishads (Advaita/Visishtadvaita/Dvaita).

9. Quick Revision Sheet

Topic	Must-remember facts
IKS origin	Vedic period, c. 1500–500 BCE
Threats to TK	Language threat, external pressures, biopiracy (turmeric patent)
Types of TK (6)	Cultural, Artistic, Medicinal, Biodiversity/Natural Resources, Agricultural, Sacred
IKS Corpus	Literary (Sanatana Dharma, Other Dharmik, Regional) + Non-Literary (Oral traditions, 64 Kalas)
Rig Veda	10 Mandalas, 1,028 hymns, 10,552 verses; Shakala & Bashkala survive
Yajur Veda	Shukla (Mantra & Brahmana separate) / Krishna (together); 6 surviving Shakhas
Sama Veda	1,549 verses; shortest; 3 recensions – Kauthuma, Jaiminiya, Ranayaniya

Topic	Must-remember facts
Atharva Veda	730 hymns, ~6,000 mantras, 20 books; Mundaka, Mandukya, Prashna Upanishads
Vedic portions	Samhita, Brahmana, Aranyaka, Upanishad
Vedangas	Siksha, Chandas, Vyakarana, Nirukta, Jyotisha, Kalpa
Key authors	Panini (Ashtadhyayi), Yaska (Nighantu), Patanjali (Mahabhashya / Yoga Sutras), Varahamihira (Pancha Siddhantika), Bhaskara Acharya (Panchanga)
Gayatri / Anushtup	$3 \times 8 = 24$ / $4 \times 8 = 32$ syllables
Kalpasutras (4)	Shrauta, Grihya, Dharma, Shulba
Astika / Nastika	6 accept Vedas / 3 reject Vedas